Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use multiplication notation.			
model multiplication of whole numbers.			
multiply whole numbers.			
translate word phrases to math notation.			
multiply whole numbers in applications.			

ⓑ *On a scale of 1–10, how would you rate your mastery of this section in light of your responses on the checklist? How can you improve this?*

1.5 Divide Whole Numbers

Learning Objectives

By the end of this section, you will be able to:

- Use division notation
- Model division of whole numbers
- Divide whole numbers
- Translate word phrases to math notation
- Divide whole numbers in applications

☑ **BE PREPARED** 1.7 Before you get started, take this readiness quiz.

Multiply: $27 \cdot 3$.
If you missed this problem, review Example 1.44.

☑ **BE PREPARED** 1.8 Subtract: $43 - 26$.
If you missed this problem, review Example 1.32

☑ **BE PREPARED** 1.9 Multiply: $62(87)$.
If you missed this problem, review Example 1.45.

Use Division Notation

So far we have explored addition, subtraction, and multiplication. Now let's consider division. Suppose you have the 12 cookies in Figure 1.13 and want to package them in bags with 4 cookies in each bag. How many bags would we need?



Figure 1.13

You might put 4 cookies in first bag, 4 in the second bag, and so on until you run out of cookies. Doing it this way, you would fill 3 bags.



In other words, starting with the 12 cookies, you would take away, or subtract, 4 cookies at a time. Division is a way to represent repeated subtraction just as multiplication represents repeated addition.

Instead of subtracting 4 repeatedly, we can write

$$12 \div 4$$

We read this as *twelve divided by four* and the result is the **quotient** of 12 and 4. The quotient is 3 because we can subtract 4 from 12 exactly 3 times. We call the number being divided the **dividend** and the number dividing it the **divisor**. In this case, the dividend is 12 and the divisor is 4.

In the past you may have used the notation $4\overline{)12}$, but this division also can be written as $12 \div 4,\ 12/4,\ \frac{12}{4}$. In each case the 12 is the dividend and the 4 is the divisor.

Operation Symbols for Division

To represent and describe division, we can use symbols and words.

Operation	Notation	Expression	Read as	Result
Division	$\div$ $\frac{a}{b}$ $b\overline{)a}$ a/b	$12 \div 4$ $\frac{12}{4}$ $4\overline{)12}$ $12/4$	Twelve divided by four	the quotient of 12 and 4

Division is performed on two numbers at a time. When translating from math notation to English words, or English words to math notation, look for the words *of* and *and* to identify the numbers.

EXAMPLE 1.56

Translate from math notation to words.

ⓐ $64 \div 8$ ⓑ $\frac{42}{7}$ ⓒ $4\overline{)28}$

Solution

ⓐ We read this as *sixty-four divided by eight* and the result is *the quotient of sixty-four and eight*.

ⓑ We read this as *forty-two divided by seven* and the result is *the quotient of forty-two and seven*.

ⓒ We read this as *twenty-eight divided by four* and the result is *the quotient of twenty-eight and four*.

TRY IT 1.111 Translate from math notation to words:

ⓐ $84 \div 7$ ⓑ $\frac{18}{6}$ ⓒ $8\overline{)24}$

TRY IT 1.112 Translate from math notation to words:

ⓐ $72 \div 9$ ⓑ $\frac{21}{3}$ ⓒ $6\overline{)54}$

Model Division of Whole Numbers

As we did with multiplication, we will model division using counters. The operation of division helps us organize items into equal groups as we start with the number of items in the dividend and subtract the number in the divisor repeatedly.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Math Worksheets activity "Model Division of Whole Numbers" will help you develop a better understanding of dividing whole numbers.

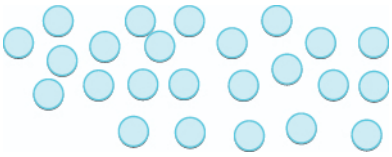
EXAMPLE 1.57

Model the division: $24 \div 8$.

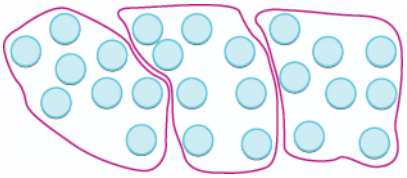
Solution

To find the quotient $24 \div 8$, we want to know how many groups of 8 are in 24.

Model the dividend. Start with 24 counters.



The divisor tell us the number of counters we want in each group. Form groups of 8 counters.



Count the number of groups. There are 3 groups.

$24 \div 8 = 3$

TRY IT 1.113 Model: $24 \div 6$.

TRY IT 1.114 Model: $42 \div 7$.

Divide Whole Numbers

We said that addition and subtraction are inverse operations because one undoes the other. Similarly, division is the inverse operation of multiplication. We know $12 \div 4 = 3$ because $3 \cdot 4 = 12$. Knowing all the multiplication number facts is very important when doing division.

We check our answer to division by multiplying the quotient by the divisor to determine if it equals the dividend. In Example 1.57, we know $24 \div 8 = 3$ is correct because $3 \cdot 8 = 24$.

EXAMPLE 1.58

Divide. Then check by multiplying. (a) $42 \div 6$ (b) $\frac{72}{9}$ (c) $7\overline{)63}$

Solution

(a)		(b)		(c)	
	$42 \div 6$		$\frac{72}{9}$		$7\overline{)63}$
Divide 42 by 6.	7	Divide 72 by 9.	8	Divide 63 by 7.	9
Check by multiplying. $7 \cdot 6$		Check by multiplying. $8 \cdot 9$		Check by multiplying. $9 \cdot 7$	
42✓		72✓		63✓	

TRY IT 1.115 Divide. Then check by multiplying:

ⓐ $54 \div 6$ ⓑ $\frac{27}{9}$

TRY IT 1.116 Divide. Then check by multiplying:

ⓐ $\frac{36}{9}$ ⓑ $8\overline{)40}$

What is the quotient when you divide a number by itself?

$$\frac{15}{15} = 1 \text{ because } 1 \cdot 15 = 15$$

Dividing any number (except 0) by itself produces a quotient of 1. Also, any number divided by 1 produces a quotient of the number. These two ideas are stated in the Division Properties of One.

Division Properties of One

Any number (except 0) divided by itself is one.	$a \div a = 1$
Any number divided by one is the same number.	$a \div 1 = a$

Table 1.6

EXAMPLE 1.59

Divide. Then check by multiplying:

ⓐ $11 \div 11$ ⓑ $\frac{19}{1}$ ⓒ $1\overline{)7}$

Solution

ⓐ

	$11 \div 11$
A number divided by itself is 1.	1
Check by multiplying. $1 \cdot 11$	
11✓	

ⓑ

	$\frac{19}{1}$
A number divided by 1 equals itself.	19
Check by multiplying. $19 \cdot 1$	
19✓	

ⓒ	
	$1\overline{)7}$
A number divided by 1 equals itself.	7
Check by multiplying. $7 \cdot 1$	
7✓	

TRY IT 1.117 Divide. Then check by multiplying:

ⓐ $14 \div 14$ ⓑ $\frac{27}{1}$

TRY IT 1.118 Divide. Then check by multiplying:

ⓐ $\frac{16}{1}$ ⓑ $1\overline{)4}$

Suppose we have $\$0$, and want to divide it among 3 people. How much would each person get? Each person would get $\$0$. Zero divided by any number is 0.

Now suppose that we want to divide $\$10$ by 0. That means we would want to find a number that we multiply by 0 to get 10. This cannot happen because 0 times any number is 0. Division by zero is said to be *undefined*.

These two ideas make up the Division Properties of Zero.

Division Properties of Zero

Zero divided by any number is 0.	$0 \div a = 0$
Dividing a number by zero is undefined.	$a \div 0$ undefined

Table 1.7

Another way to explain why division by zero is undefined is to remember that division is really repeated subtraction. How many times can we take away 0 from 10? Because subtracting 0 will never change the total, we will never get an answer. So we cannot divide a number by 0.

EXAMPLE 1.60

Divide. Check by multiplying: ⓐ $0 \div 3$ ⓑ $10/0$.

Solution

ⓐ

	$0 \div 3$
Zero divided by any number is zero.	0
Check by multiplying. $0 \cdot 3$	
0✓	

ⓑ

	$10/0$
Division by zero is undefined.	undefined

TRY IT 1.119 Divide. Then check by multiplying:

ⓐ $0 \div 2$ ⓑ $17/0$

TRY IT 1.120 Divide. Then check by multiplying:

ⓐ $0 \div 6$ ⓑ $13/0$

When the divisor or the dividend has more than one digit, it is usually easier to use the $4\overline{)12}$ notation. This process is called long division. Let's work through the process by dividing 78 by 3.

Divide the first digit of the dividend, 7, by the divisor, 3.	
The divisor 3 can go into 7 two times since $2 \times 3 = 6$. Write the 2 above the 7 in the quotient.	$3\overline{)78}$ → $\begin{array}{r}2\\3\overline{)78}\end{array}$
Multiply the 2 in the quotient by 3 and write the product, 6, under the 7.	$\begin{array}{r}2\\3\overline{)78}\\6\end{array}$
Subtract that product from the first digit in the dividend. Subtract $7 - 6$. Write the difference, 1, under the first digit in the dividend.	$\begin{array}{r}2\\3\overline{)78}\\6\\\hline 1\end{array}$
Bring down the next digit of the dividend. Bring down the 8.	$\begin{array}{r}2\\3\overline{)78}\\6\\\hline 18\end{array}$
Divide 18 by the divisor, 3. The divisor 3 goes into 18 six times.	$\begin{array}{r}26\\3\overline{)78}\\6\\\hline 18\end{array}$
Write 6 in the quotient above the 8.	
Multiply the 6 in the quotient by the divisor and write the product, 18, under the dividend. Subtract 18 from 18.	$\begin{array}{r}26\\3\overline{)78}\\6\\\hline 18\\18\\\hline 0\end{array}$

We would repeat the process until there are no more digits in the dividend to bring down. In this problem, there are no more digits to bring down, so the division is finished.

$$\text{So } 78 \div 3 = 26.$$

Check by multiplying the quotient times the divisor to get the dividend. Multiply 26×3 to make sure that product equals the dividend, 78.

$$\begin{array}{r} {}^{1}26 \\ \times 3 \\ \hline 78\checkmark \end{array}$$

It does, so our answer is correct.

HOW TO

Divide whole numbers.

Step 1. Divide the first digit of the dividend by the divisor.
If the divisor is larger than the first digit of the dividend, divide the first two digits of the dividend by the divisor, and so on.
Step 2. Write the quotient above the dividend.
Step 3. Multiply the quotient by the divisor and write the product under the dividend.
Step 4. Subtract that product from the dividend.
Step 5. Bring down the next digit of the dividend.
Step 6. Repeat from Step 1 until there are no more digits in the dividend to bring down.
Step 7. Check by multiplying the quotient times the divisor.

EXAMPLE 1.61

Divide $2{,}596 \div 4$. Check by multiplying:

Solution

Let's rewrite the problem to set it up for long division.	$4\overline{)2596}$
Divide the first digit of the dividend, 2, by the divisor, 4.	$4\overline{)2596}$
Since 4 does not go into 2, we use the first two digits of the dividend and divide 25 by 4. The divisor 4 goes into 25 six times.	
We write the 6 in the quotient above the 5.	$\begin{array}{r} 6 \\ 4\overline{)2596} \end{array}$
Multiply the 6 in the quotient by the divisor 4 and write the product, 24, under the first two digits in the dividend.	$\begin{array}{r} 6 \\ 4\overline{)2596} \\ \underline{24} \end{array}$
Subtract that product from the first two digits in the dividend. Subtract $25 - 24$. Write the difference, 1, under the second digit in the dividend.	$\begin{array}{r} 6 \\ 4\overline{)2596} \\ \underline{24} \\ 1 \end{array}$

Now bring down the 9 and repeat these steps. There are 4 fours in 19. Write the 4 over the 9. Multiply the 4 by 4 and subtract this product from 19.	$\begin{array}{r} 64 \\ 4\overline{)2596} \\ 24 \\ \hline 19 \\ 16 \\ \hline 3 \end{array}$
Bring down the 6 and repeat these steps. There are 9 fours in 36. Write the 9 over the 6. Multiply the 9 by 4 and subtract this product from 36.	$\begin{array}{r} 649 \\ 4\overline{)2596} \\ 24 \\ \hline 19 \\ 16 \\ \hline 36 \\ 36 \\ \hline 0 \end{array}$
So $2{,}596 \div 4 = 649$.	
Check by multiplying. $\begin{array}{r} {\scriptstyle 1\ 3} \\ 649 \\ \times \quad 4 \\ \hline 2{,}596 \checkmark \end{array}$	

It equals the dividend, so our answer is correct.

TRY IT 1.121 Divide. Then check by multiplying: $2{,}636 \div 4$

TRY IT 1.122 Divide. Then check by multiplying: $2{,}716 \div 4$

EXAMPLE 1.62

Divide $4{,}506 \div 6$. Check by multiplying:

Solution

Let's rewrite the problem to set it up for long division.	$6\overline{)4506}$
First we try to divide 6 into 4.	$6\overline{)4506}$
Since that won't work, we try 6 into 45. There are 7 sixes in 45. We write the 7 over the 5.	$\begin{array}{r} 7 \\ 6\overline{)4506} \end{array}$
Multiply the 7 by 6 and subtract this product from 45.	$\begin{array}{r} 7 \\ 6\overline{)4506} \\ 42 \\ \hline 3 \end{array}$

Now bring down the 0 and repeat these steps. There are 5 sixes in 30. Write the 5 over the 0. Multiply the 5 by 6 and subtract this product from 30.	$\begin{array}{r} 75 \\ 6\overline{)4506} \\ 42 \\ \hline 30 \\ 30 \\ \hline 0 \end{array}$
Now bring down the 6 and repeat these steps. There is 1 six in 6. Write the 1 over the 6. Multiply 1 by 6 and subtract this product from 6.	$\begin{array}{r} 751 \\ 6\overline{)4506} \\ 42 \\ \hline 30 \\ 30 \\ \hline 06 \\ 6 \\ \hline 0 \end{array}$
Check by multiplying. $\begin{array}{r} {\scriptstyle 3}\,751 \\ \times\quad 6 \\ \hline 4{,}506 \checkmark \end{array}$	

It equals the dividend, so our answer is correct.

TRY IT 1.123 Divide. Then check by multiplying: $4{,}305 \div 5$.

TRY IT 1.124 Divide. Then check by multiplying: $3{,}906 \div 6$.

EXAMPLE 1.63

Divide $7{,}263 \div 9$. Check by multiplying.

Solution

Let's rewrite the problem to set it up for long division.	$9\overline{)7263}$
First we try to divide 9 into 7.	$9\overline{)7263}$
Since that won't work, we try 9 into 72. There are 8 nines in 72. We write the 8 over the 2.	$\begin{array}{r} 8 \\ 9\overline{)7263} \end{array}$
Multiply the 8 by 9 and subtract this product from 72.	$\begin{array}{r} 8 \\ 9\overline{)7263} \\ 72 \\ \hline 0 \end{array}$
Now bring down the 6 and repeat these steps. There are 0 nines in 6. Write the 0 over the 6. Multiply the 0 by 9 and subtract this product from 6.	$\begin{array}{r} 80 \\ 9\overline{)7263} \\ 72 \\ \hline 06 \\ 0 \\ \hline 6 \end{array}$

Now bring down the 3 and repeat these steps. There are 7 nines in 63. Write the 7 over the 3. Multiply the 7 by 9 and subtract this product from 63.

$$\begin{array}{r} 807 \\ 9\overline{)7263} \\ \underline{72} \\ 06 \\ \underline{0} \\ 63 \\ \underline{63} \\ 0 \end{array}$$

Check by multiplying.

$$\begin{array}{r} {}^{6}\,807 \\ \times\quad 9 \\ \hline 7{,}263\ \checkmark \end{array}$$

It equals the dividend, so our answer is correct.

TRY IT 1.125 Divide. Then check by multiplying: $4{,}928 \div 7$.

TRY IT 1.126 Divide. Then check by multiplying: $5{,}663 \div 7$.

So far all the division problems have worked out evenly. For example, if we had 24 cookies and wanted to make bags of 8 cookies, we would have 3 bags. But what if there were 28 cookies and we wanted to make bags of 8? Start with the 28 cookies as shown in Figure 1.14.



Figure 1.14

Try to put the cookies in groups of eight as in Figure 1.15.

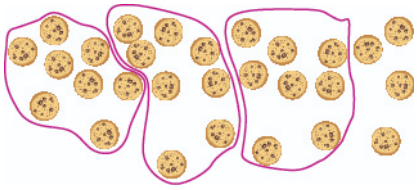


Figure 1.15

There are 3 groups of eight cookies, and 4 cookies left over. We call the 4 cookies that are left over the remainder and show it by writing R4 next to the 3. (The R stands for remainder.)

To check this division we multiply 3 times 8 to get 24, and then add the remainder of 4.

$$\begin{array}{r} 3 \\ \times 8 \\ \hline 24 \\ +4 \\ \hline 28 \end{array}$$

EXAMPLE 1.64

Divide $1{,}439 \div 4$. Check by multiplying.

Solution

Let's rewrite the problem to set it up for long division.	$4\overline{)1439}$
First we try to divide 4 into 1. Since that won't work, we try 4 into 14. There are 3 fours in 14. We write the 3 over the 4.	$\begin{array}{r} 3 \\ 4\overline{)1439} \end{array}$
Multiply the 3 by 4 and subtract this product from 14.	$\begin{array}{r} 3 \\ 4\overline{)1439} \\ \underline{12} \\ 2 \end{array}$
Now bring down the 3 and repeat these steps. There are 5 fours in 23. Write the 5 over the 3. Multiply the 5 by 4 and subtract this product from 23.	$\begin{array}{r} 35 \\ 4\overline{)1439} \\ \underline{12}\downarrow \\ 23 \\ \underline{20} \\ 3 \end{array}$
Now bring down the 9 and repeat these steps. There are 9 fours in 39. Write the 9 over the 9. Multiply the 9 by 4 and subtract this product from 39. There are no more numbers to bring down, so we are done. The remainder is 3.	$\begin{array}{r} 359\text{R}3 \\ 4\overline{)1439} \\ \underline{12} \\ 23 \\ \underline{20}\downarrow \\ 39 \\ \underline{36} \\ 3 \end{array}$
Check by multiplying. $\begin{array}{rl} \overset{2}{3}\overset{3}{5}9 & \text{quotient} \\ \times\ 4 & \text{divisor} \\ \hline 1{,}436 & \\ +\ 3 & \text{remainder} \\ \hline 1{,}439\checkmark & \end{array}$	

So $1{,}439 \div 4$ is 359 with a remainder of 3. Our answer is correct.

TRY IT 1.127 Divide. Then check by multiplying: $3{,}812 \div 8$.

TRY IT 1.128 Divide. Then check by multiplying: $4{,}319 \div 8$.

EXAMPLE 1.65

Divide and then check by multiplying: $1{,}461 \div 13$.

Solution

Let's rewrite the problem to set it up for long division.	$13\overline{)1{,}461}$
First we try to divide 13 into 1. Since that won't work, we try 13 into 14. There is 1 thirteen in 14. We write the 1 over the 4.	$\begin{array}{r} 1 \\ 13\overline{)1461} \end{array}$

Multiply the 1 by 13 and subtract this product from 14.	$\begin{array}{r} 1 \\ 13\overline{)1461} \\ 13 \\ \hline 1 \end{array}$
Now bring down the 6 and repeat these steps. There is 1 thirteen in 16. Write the 1 over the 6. Multiply the 1 by 13 and subtract this product from 16.	$\begin{array}{r} 11 \\ 13\overline{)1461} \\ 13\downarrow \\ \hline 16 \\ 13 \\ \hline 3 \end{array}$
Now bring down the 1 and repeat these steps. There are 2 thirteens in 31. Write the 2 over the 1. Multiply the 2 by 13 and subtract this product from 31. There are no more numbers to bring down, so we are done. The remainder is 5. $1{,}462 \div 13$ is 112 with a remainder of 5.	$\begin{array}{r} 112\text{R}5 \\ 13\overline{)1461} \\ 13\downarrow \\ \hline 16\downarrow \\ 13 \\ \hline 31 \\ 26 \\ \hline 5 \end{array}$
Check by multiplying. $\begin{array}{rl} 112 & \text{quotient} \\ \times\ 13 & \text{divisor} \\ \hline 336 & \\ 1{,}120 & \\ +\ \ 5 & \text{remainder} \\ \hline 1{,}461 & \checkmark \end{array}$	

Our answer is correct.

TRY IT 1.129 Divide. Then check by multiplying: $1{,}493 \div 13$.

TRY IT 1.130 Divide. Then check by multiplying: $1{,}461 \div 12$.

EXAMPLE 1.66

Divide and check by multiplying: $74{,}521 \div 241$.

Solution

Let's rewrite the problem to set it up for long division.	$241\overline{)74{,}521}$
First we try to divide 241 into 7. Since that won't work, we try 241 into 74. That still won't work, so we try 241 into 745. Since 2 divides into 7 three times, we try 3. Since $3 \times 241 = 723$, we write the 3 over the 5 in 745. Note that 4 would be too large because $4 \times 241 = 964$, which is greater than 745.	
Multiply the 3 by 241 and subtract this product from 745.	$\begin{array}{r} 3 \\ 241\overline{)74521} \\ 723 \\ \hline 22 \end{array}$

Now bring down the 2 and repeat these steps. 241 does not divide into 222. We write a 0 over the 2 as a placeholder and then continue.	$$\begin{array}{r} 30 \\ 241\overline{)74521} \\ \underline{723} \\ 222 \end{array}$$
Now bring down the 1 and repeat these steps. Try 9. Since $9 \times 241 = 2{,}169$, we write the 9 over the 1. Multiply the 9 by 241 and subtract this product from 2,221.	$$\begin{array}{r} 309\text{ R}52 \\ 241\overline{)74521} \\ \underline{723} \\ 2221 \\ \underline{2169} \\ 52 \end{array}$$
There are no more numbers to bring down, so we are finished. The remainder is 52. So $74{,}521 \div 241$ is 309 with a remainder of 52.	
Check by multiplying. $$\begin{array}{rl} {\scriptstyle 3}\,309 & \text{quotient} \\ \times\ 241 & \text{divisor} \\ \hline 309 & \\ 12{,}360 & \\ \underline{61{,}800} & \\ 74{,}469 & \\ +\ \ 52 & \text{remainder} \\ \hline 74{,}521 & \checkmark \end{array}$$	

Sometimes it might not be obvious how many times the divisor goes into digits of the dividend. We will have to guess and check numbers to find the greatest number that goes into the digits without exceeding them.

TRY IT 1.131 Divide. Then check by multiplying: $78{,}641 \div 256$.

TRY IT 1.132 Divide. Then check by multiplying: $76{,}461 \div 248$.

Translate Word Phrases to Math Notation

Earlier in this section, we translated math notation for division into words. Now we'll translate word phrases into math notation. Some of the words that indicate division are given in Table 1.8.

Operation	Word Phrase	Example	Expression
Division	divided by quotient of divided into	12 divided by 4 the quotient of 12 and 4 4 divided into 12	$12 \div 4$ $\frac{12}{4}$ $12/4$ $4\overline{)12}$

Table 1.8

EXAMPLE 1.67

Translate and simplify: the quotient of 51 and 17.

Solution

The word *quotient* tells us to divide.

	the quotient of 51 and 17
Translate.	$51 \div 17$
Divide.	3

We could just as correctly have translated *the quotient of 51 and 17* using the notation $17\overline{)51}$ or $\frac{51}{17}$.

TRY IT 1.133 Translate and simplify: the quotient of 91 and 13.

TRY IT 1.134 Translate and simplify: the quotient of 52 and 13.

Divide Whole Numbers in Applications

We will use the same strategy we used in previous sections to solve applications. First, we determine what we are looking for. Then we write a phrase that gives the information to find it. We then translate the phrase into math notation and simplify it to get the answer. Finally, we write a sentence to answer the question.

EXAMPLE 1.68

Cecelia bought a 160-ounce box of oatmeal at the big box store. She wants to divide the 160 ounces of oatmeal into 8-ounce servings. She will put each serving into a plastic bag so she can take one bag to work each day. How many servings will she get from the big box?

Solution

We are asked to find the how many servings she will get from the big box.

Write a phrase.	160 ounces divided by 8 ounces
Translate to math notation.	$160 \div 8$
Simplify by dividing.	20
Write a sentence to answer the question.	Cecelia will get 20 servings from the big box.

TRY IT 1.135 Marcus is setting out animal crackers for snacks at the preschool. He wants to put 9 crackers in each cup. One box of animal crackers contains 135 crackers. How many cups can he fill from one box of crackers?

TRY IT 1.136 Andrea is making bows for the girls in her dance class to wear at the recital. Each bow takes 4 feet of ribbon, and 36 feet of ribbon are on one spool. How many bows can Andrea make from one spool of ribbon?

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Dividing Whole Numbers (https://www.openstax.org/l/24divwhlnum)
Dividing Whole Numbers No Remainder (https://www.openstax.org/l/24divnumnorem)
Dividing Whole Numbers With Remainder (https://www.openstax.org/l/24divnumwrem)

SECTION 1.5 EXERCISES

Practice Makes Perfect

Use Division Notation

In the following exercises, translate from math notation to words.

343. $54 \div 9$

344. $\frac{56}{7}$

345. $\frac{32}{8}$

346. $6\overline{)42}$

347. $48 \div 6$

348. $\frac{63}{9}$

349. $7\overline{)63}$

350. $72 \div 8$

Model Division of Whole Numbers

In the following exercises, model the division.

351. $15 \div 5$

352. $10 \div 5$

353. $\frac{14}{7}$

354. $\frac{18}{6}$

355. $4\overline{)20}$

356. $3\overline{)15}$

357. $24 \div 6$

358. $16 \div 4$

Divide Whole Numbers

In the following exercises, divide. Then check by multiplying.

359. $18 \div 2$

360. $14 \div 2$

361. $\frac{27}{3}$

362. $\frac{30}{3}$

363. $4\overline{)28}$

364. $4\overline{)36}$

365. $\frac{45}{5}$

366. $\frac{35}{5}$

367. $72/8$

368. $8\overline{)64}$

369. $\frac{35}{7}$

370. $42 \div 7$

371. $15\overline{)15}$

372. $12\overline{)12}$

373. $43 \div 43$

374. $37 \div 37$

375. $\frac{23}{1}$

376. $\frac{29}{1}$

377. $19 \div 1$

378. $17 \div 1$

379. $0 \div 4$

380. $0 \div 8$

381. $\frac{5}{0}$

382. $\frac{9}{0}$

383. $\frac{26}{0}$

384. $\frac{32}{0}$

385. $12\overline{)0}$

386. $16\overline{)0}$

387. $72 \div 3$

388. $57 \div 3$

389. $\frac{96}{8}$

390. $\frac{78}{6}$

391. $5\overline{)465}$

392. $4\overline{)528}$

393. $924 \div 7$

394. $861 \div 7$

395. $\frac{5{,}226}{6}$

396. $\frac{3{,}776}{8}$

397. $4\overline{)31{,}324}$

398. $5\overline{)46{,}855}$

399. $7{,}209 \div 3$

400. $4{,}806 \div 3$

401. $5{,}406 \div 6$

402. $3{,}208 \div 4$

403. $4\overline{)2{,}816}$

404. $6\overline{)3{,}624}$

405. $\frac{91{,}881}{9}$

406. $\frac{83{,}256}{8}$

407. $2{,}470 \div 7$

408. $3{,}741 \div 7$

409. $8\overline{)55{,}305}$

410. $9\overline{)51{,}492}$

411. $\frac{431{,}174}{5}$

412. $\frac{297{,}277}{4}$

413. $130{,}016 \div 3$

414. $105{,}609 \div 2$

415. $15\overline{)5{,}735}$

416. $\frac{4{,}933}{21}$

417. $56{,}883 \div 67$

418. $43{,}725/75$

419. $\frac{30{,}144}{314}$

420. $26{,}145 \div 415$

421. $273\overline{)542{,}195}$

422. $816{,}243 \div 462$

Mixed Practice

In the following exercises, simplify.

423. $15\,(204)$

424. $74 \cdot 391$

425. $256 - 184$

426. $305 - 262$

427. $719 + 341$

428. $647 + 528$

429. $25\overline{)875}$

430. $1104 \div 23$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate and simplify.

431. the quotient of 45 and 15

432. the quotient of 64 and 16

433. the quotient of 288 and 24

434. the quotient of 256 and 32

Divide Whole Numbers in Applications

In the following exercises, solve.

435. **Trail mix** Ric bought 64 ounces of trail mix. He wants to divide it into small bags, with 2 ounces of trail mix in each bag. How many bags can Ric fill?

436. **Crackers** Evie bought a 42 ounce box of crackers. She wants to divide it into bags with 3 ounces of crackers in each bag. How many bags can Evie fill?

437. **Astronomy class** There are 125 students in an astronomy class. The professor assigns them into groups of 5. How many groups of students are there?

438. **Flower shop** Melissa's flower shop got a shipment of 152 roses. She wants to make bouquets of 8 roses each. How many bouquets can Melissa make?

439. **Baking** One roll of plastic wrap is 48 feet long. Marta uses 3 feet of plastic wrap to wrap each cake she bakes. How many cakes can she wrap from one roll?

440. **Dental floss** One package of dental floss is 54 feet long. Brian uses 2 feet of dental floss every day. How many days will one package of dental floss last Brian?

Mixed Practice

In the following exercises, solve.

441. **Miles per gallon** Susana's hybrid car gets 45 miles per gallon. Her son's truck gets 17 miles per gallon. What is the difference in miles per gallon between Susana's car and her son's truck?

442. **Distance** Mayra lives 53 miles from her mother's house and 71 miles from her mother-in-law's house. How much farther is Mayra from her mother-in-law's house than from her mother's house?

443. **Field trip** The 45 students in a Geology class will go on a field trip, using the college's vans. Each van can hold 9 students. How many vans will they need for the field trip?

444. **Potting soil** Aki bought a 128 ounce bag of potting soil. How many 4 ounce pots can he fill from the bag?

445. **Hiking** Bill hiked 8 miles on the first day of his backpacking trip, 14 miles the second day, 11 miles the third day, and 17 miles the fourth day. What is the total number of miles Bill hiked?

446. **Reading** Last night Emily read 6 pages in her Business textbook, 26 pages in her History text, 15 pages in her Psychology text, and 9 pages in her math text. What is the total number of pages Emily read?

447. **Patients** LaVonne treats 12 patients each day in her dental office. Last week she worked 4 days. How many patients did she treat last week?

448. **Scouts** There are 14 boys in Dave's scout troop. At summer camp, each boy earned 5 merit badges. What was the total number of merit badges earned by Dave's scout troop at summer camp?

Writing Exercises

449. **Contact lenses** Jenna puts in a new pair of contact lenses every 14 days. How many pairs of contact lenses does she need for 365 days?

450. **Cat food** One bag of cat food feeds Lara's cat for 25 days. How many bags of cat food does Lara need for 365 days?

Everyday Math

451. Explain how you use the multiplication facts to help with division.

452. Oswaldo divided 300 by 8 and said his answer was 37 with a remainder of 4. How can you check to make sure he is correct?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use division notation.			
model division of whole numbers.			
divide whole numbers.			
translate word phrases to algebraic expressions.			
divide whole numbers in applications.			

ⓑ *Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?*